SAT Math Mastery · Session 2

Nonlinear Functions

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics quadratic models: vertex, axis of symmetry, maximum and minimum · writing a quadratic from given conditions · transformations and composition · exponential growth and decay · equivalent exponential forms · comparing functions · factors and zeros · reading graphs and tables · rational and radical equations · inverse functions · maximum and minimum with a constraint

Name / Date _____ _____

PART A · IN CLASS — 48 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

The amount of a medicine in a patient's body is modeled by $A(t) = 640\left(\frac{1}{2}\right)^{\frac{t}{8}}$ milligrams, where t is the number of hours after the dose. After how many hours does the model estimate exactly 40 milligrams remain?

1. Type `y=640*(1/2)^(x/8)` and `y=40` (letting x stand for t; zoom out to see both).

2. Tap the intersection: (32, 40). Its x-value is the time.

3. Type `640*(1/2)^(32/8)`: it returns 40, so the dose is at 40 milligrams after 32 hours.

Answer: 32

2. HARD · DESMOS

The function $h(t) = -16t^2 + 96t + 4$ models the height, in feet, of a ball t seconds after it is launched. The graph of $y = h(t)$ in the ty-plane has a vertex at $(3, 148)$. Which of the following is the best interpretation of the vertex in this context?

(A) The ball was launched from a height of 148 feet.

(B) The ball reached its maximum height of 148 feet 3 seconds after it was launched. ✓

(C) The ball hit the ground 148 seconds after it was launched.

(D) The ball was 3 feet above the ground 148 seconds after it was launched.

1. Type `y=-16x^2+96x+4` (letting x stand for t) and tap the top of the curve (zoom out if needed).

2. The vertex is (3, 148), the highest point because the parabola opens downward.

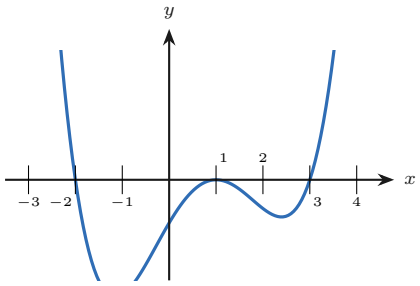
3. The first coordinate is time and the second is height, so 3 seconds after launch the ball is at its maximum height of 148 feet.

Answer: (B)

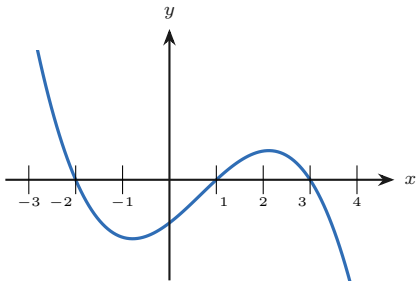
3. SAT Challenge · Desmos

Which of the following could be the graph of $y = -(x+2)(x-1)^2(x-3)$ in the xy-plane?

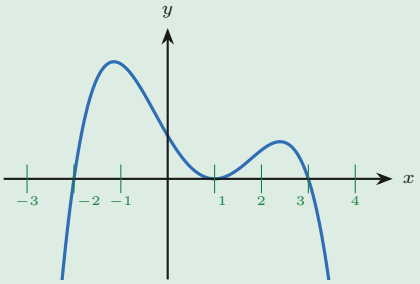
(A)



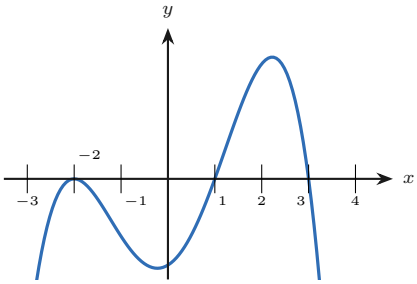
(B)



(C) ✓



(D)



1. A factor with an even power (the square $(x-1)^2$) makes the graph touch the x-axis and turn at $x = 1$. The factors $(x+2)$ and $(x-3)$ appear once, so the graph crosses there. Degree 4 with leading coefficient -1 means both ends point down.
2. Type `y=-(x+2)(x-1)^2(x-3)` : the zeros are $(-2, 0)$, $(1, 0)$ and $(3, 0)$, and the curve touches at $x = 1$ without crossing.
3. Only graph (C) touches at 1, crosses at -2 and 3, and has both ends down. Graph (D) touches at -2 instead, graph (A) has both ends up, and graph (B) crosses at 1 with ends in opposite directions.

Answer: (C)

4. Hard · Desmos

A theater sells 600 tickets when the price is \$20. For each \$1 increase in price, 20 fewer tickets are sold. Which ticket price, in dollars, gives the greatest revenue?

(A) 25 ✓

(B) 12,000

(C) 12,500

(D) 5

1. Let x be the number of \$1 increases. The price is $20 + x$ and the tickets sold are $600 - 20x$. Type `y=(20+x)(600-20x)` .
2. Tap the vertex (zoom out if needed): (5, 12500).
3. That is $x = 5$ increases of \$1. Type `20+5` : it returns 25, the price in dollars. The value 12,500 is the greatest revenue, not the price.

Answer: (A)

5. HARD · DESMOS

The cost, in dollars, of making each unit when a factory produces n units is modeled by $C(n) = 0.02n^2 - 3.2n + 210$. What is the minimum cost per unit, in dollars, according to the model?

1. Type `y=0.02x^2-3.2x+210` (letting x stand for n) and zoom out.

2. Tap the vertex (lowest point): (80, 82).

3. Type `0.02*80^2-3.2*80+210`: it returns 82, so the minimum cost per unit is 82 dollars.

Answer: 82

6. HARD · DESMOS

The quadratic function $f(x) = ax^2 + bx + c$, where a, b and c are constants, has two distinct real zeros. It is known that $f(2) = f(8)$, $f(0) < 0$ and $f(5) > 0$. Consider the following statements.

I. $b > 0$
II. The product of the two zeros is greater than 25.
III. Both zeros are positive.

Which of the statements must be true?

(A) I, II, and III

(B) I only

(C) II and III only

(D) I and III only ✓

1. $f(2) = f(8)$ puts the axis at $x = 5$. With two zeros and $f(5) > 0$ the parabola opens downward. Try an example that fits: type `y=-(x-5)^2+9`.

2. Tap the points: zeros (2, 0) and (8, 0) (both positive), y-intercept (0, −16) (so $f(0) < 0$), vertex (5, 9) (so $f(5) > 0$). Statement III holds.

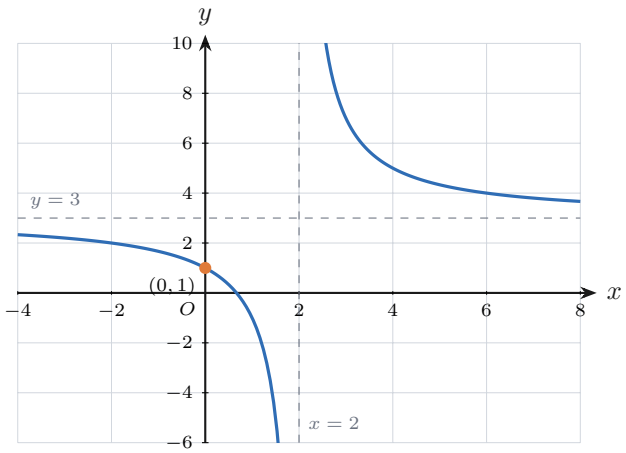
3. Type `y=-x^2+10x-16`: the same graph, so $b = 10 > 0$ and statement I holds.

4. The zeros multiply to $2 \cdot 8 = 16$, not more than 25 (in general they are $5 \pm d$ with $d < 5$, product $25 - d^2 < 25$), so II fails. Statements I and III only.

Answer: (D)

7. HARD · DESMOS

The graph of the rational function $f(x) = \dfrac{ax+b}{x-c}$ is shown, where a, b and c are constants. The dashed lines are the asymptotes $x = 2$ and $y = 3$, and the graph passes through the point $(0, 1)$. What is the value of $f(5)$? Enter your answer as a fraction.



1. The horizontal asymptote $y = 3$ gives $a = 3$ and the vertical asymptote $x = 2$ gives $c = 2$. Type `f(x)=(3x+b)/(x-2)` (accept the slider for b) and plot `(0,1)`.
2. Slide b until the curve passes through the point: $b = -2$.
3. Type `f(5)`: it returns 4.3333, which is $\dfrac{13}{3}$.

Answer: $\dfrac{13}{3}$

8. HARD · DESMOS

The quadratic function f has its vertex at $(3, -2)$, and $f(5) = 6$. The function g is defined by $g(x) = \dfrac{3}{4} f\left(\dfrac{1}{2}x - 1\right)$. The graph of $y = f(x)$ has y-intercept $(0, p)$ and the graph of $y = g(x)$ has y-intercept $(0, q)$. What is the value of $q - p$?

(A) $\dfrac{45}{2}$

(B) $-\dfrac{23}{2}$

(C) $\dfrac{13}{2}$ ✓

(D) -4

1. Type `f(x)=a(x-3)^2-2` (accept the slider for a) and plot the point `(5,6)`.
2. Slide a until the curve passes through the point: $a = 2$. Then $p = f(0) = 16$.
3. Type `g(x)=3/4*f(x/2-1)` and then `g(0)-f(0)`: it returns 6.5, which is $\dfrac{13}{2}$.

Answer: (C)

9. HARD · DESMOS

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p, $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p?

1. Subtract the two equations: $2^p - 2^{-p} = 9 - \frac{21}{4} = \frac{15}{4}$. Type `y=2^x-2^(-x)` and `y=15/4`.

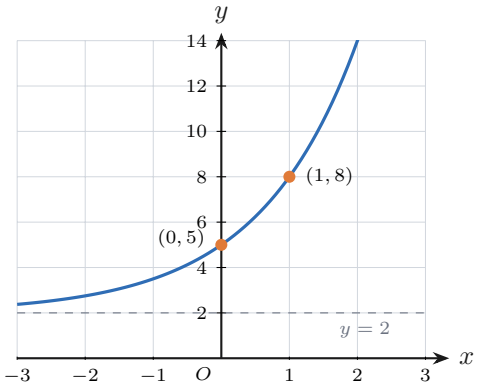
2. Tap the intersection: $(2,\ 3.75)$, so $p = 2$.

3. Then $k = 9 - 2^2 = 5$. Check: type `2^(-2)+5`: it returns 5.25, which is $\frac{21}{4}$.

Answer: 2

10. HARD · DESMOS

The graph of the exponential function f is shown. The dashed line $y = 2$ is the horizontal asymptote, and the graph passes through the points $(0, 5)$ and $(1, 8)$. Which of the following could be an equation for f?



(A) $f(x) = 2^x + 4$

(B) $f(x) = 3 \cdot 2^x + 2$ ✓

(C) $f(x) = 2 \cdot 3^x + 3$

(D) $f(x) = 5 \cdot 2^x$

1. The asymptote $y = 2$ is the vertical shift. Type `f(x)=a*b^x+2` (accept the sliders for a and b) and plot `(0,5)` and `(1,8)`.

2. At $x = 0$: type `5-2`: it returns 3, so $a = 3$. At $x = 1$: type `(8-2)/3`: it returns 2, so $b = 2$.

3. Type each option as its own line: only (B) passes through both points. For example the second option `y=2*3^x+3` reaches 9 at $x = 1$, not 8.

Answer: (B)

11. HARD · DESMOS

The altitude h, in meters, of a small rocket t seconds after launch is modeled by $h(t) = at^2 + bt + c$, where a, b, and c are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

1. The altitude points are $(0, 3)$, $(2, 43)$, $(5, 28)$. Open a table and enter $x_1 = 0, 2, 5$ and $y_1 = 3, 43, 28$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = -5$, $b = 30$, $c = 3$.

2. So $h(t) = -5t^2 + 30t + 3$.

3. Type `-5*3^2+30*3+3`: it returns 48, the altitude 3 seconds after launch in meters.

Answer: 48

12. HARD · DESMOS

The quadratic function f has zeros at $x = -2$ and $x = 5$, and $f(1) = -24$. The function h is defined by $h(x) = f(x + 2) - 3f(-x)$. What is the value of $h(1)$?

1. The zeros are -2 and 5: type `f(x)=a(x+2)(x-5)` (slider a) and plot `(1,-24)`.

2. Slide a until the curve passes through the point: $a = 2$.

3. Type `h(x)=f(x+2)-3f(-x)` and then `h(1)`: it returns 16.

Answer: 16

13. HARD · DESMOS

The quadratic function f has an axis of symmetry at $x = 2$, and its graph passes through $(3, -3)$ and $(5, 13)$. The functions g and h are defined by $g(x) = \dfrac{x - 6}{2}$ and $h(x) = 3 - f(g(x))$. What is the y-coordinate of the y-intercept of the graph of $y = h(x)$?

1. The axis is $x = 2$, so $f(1) = f(3) = -3$. Open a table and enter $x_1 = 1, 3, 5$ and $y_1 = -3, -3, 13$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = 2$, $b = -8$, $c = 3$.

2. Type `f(x)=2x^2-8x+3`, `g(x)=(x-6)/2` and `h(x)=3-f(g(x))`.

3. Type `h(0)`: it returns -42, the y-intercept's y-coordinate.

Answer: -42

14. SAT Challenge · Desmos

The function f is defined by $f(x) = a \cdot b^x$, where a and b are positive constants. Some values of f are shown in the table below.

x	1	4
$f(x)$	54	16

What is the value of $f(-2)$?

1. From $x = 1$ to $x = 4$ is 3 steps, so $b^3 = \dfrac{16}{54}$. Type `(16/54)^(1/3)` : it returns 0.6667, so $b = \dfrac{2}{3}$.

2. Then $a \cdot \dfrac{2}{3} = 54$. Type `54/(2/3)` : it returns 81, so $a = 81$.

3. Type `81*(2/3)^(-2)` : it returns 182.25, so $f(-2) = 182.25$.

4. Check: type `81*(2/3)^4` : it returns 16, which matches $f(4) = 16$.

Answer: 182.25

15. Hard · Desmos

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x, $f(x)$ decreases by p%. What is the value of p?

1. $f(3)$ and $f(1)$ are two steps apart, so $b^2 = \dfrac{729}{1296}$. Type `sqrt(729/1296)` : it returns 0.75, so $b = 0.75$.

2. Each increase of 1 multiplies f by 0.75: type `(1-0.75)*100` : it returns 25, a decrease of 25%.

3. Check: type `1296/0.75^4` : it returns 4096, so $c = 4096$. Then type `4096*0.75^6` : it returns 729, which is $f(3)$.

Answer: 25

16. Hard · Desmos

The value V, in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18{,}000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

(A) It decreases by 70% each year.

(B) It decreases by 51% each year.

(C) It decreases by 24.5% each year.

(D) It decreases by 30% each year. ✓

1. One year multiplies the value by $0.49^{1/2}$. Type `0.49^(1/2)` : it returns 0.7.

2. The yearly factor is 0.7, so the loss is $(1 - 0.7) \cdot 100$ percent. Type `(1-0.7)*100` : it returns 30.

3. Check: type `18000*0.49^(1/2)` : it returns 12600, which is $0.7 \cdot 18{,}000$, a drop of 5,400 from 18,000.

Answer: (D)

17. HARD · NO CALCULATOR NEEDED

A sample of a radioactive isotope has a mass of 80 milligrams. The half-life of the isotope is 12 days. Which of the following functions gives the mass, in milligrams, of the sample w weeks after the sample is measured?

(A) $m(w) = 80\left(\frac{1}{2}\right)^{7w/12}$ ✓

(B) $m(w) = 80\left(\frac{1}{2}\right)^{12w/7}$

(C) $m(w) = 80\left(\frac{1}{2}\right)^{84w}$

(D) $m(w) = 80\left(\frac{1}{2}\right)^{w/12}$

1. w weeks is $7w$ days.

2. The number of half-lives that pass is the number of days divided by 12: $\frac{7w}{12}$.

3. So the mass is $80\left(\frac{1}{2}\right)^{\frac{7w}{12}}$. Check: after 12 days, which is $w = \frac{12}{7}$, the exponent is $\frac{7 \cdot 12/7}{12} = 1$ and the mass is 40, half of 80. That is option (A).

Answer: (A)

18. HARD · DESMOS

In the xy-plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x-coordinate 1 and one with x-coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

1. On the line $y = 5 - x$: at $x = 1$, $y = 4$; at $x = 4$, $y = 1$. So p passes through $(1, 4)$, $(4, 1)$ and $(0, -3)$.

2. Open a table and enter $x_1 = 0, 1, 4$ and $y_1 = -3, 4, 1$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = -2$, $b = 9$, $c = -3$.

3. Type `-2*6^2+9*6-3`: it returns -21, so $p(6) = -21$.

Answer: -21

19. HARD · DESMOS

The function f is defined by $f(x) = ab^x + c$, where a, b, and c are constants and $b > 1$. The values $f(2) = 14$, $f(4) = 20$, and $f(6) = 32$ are known. What is the value of c?

1. The differences are $20 - 14 = 6$ and $32 - 20 = 12$, and the second is b^2 times the first. Type `12/6`: it returns 2, so $b^2 = 2$ and $b = \sqrt{2}$.

2. Then $ab^2(b^2 - 1) = 6$ becomes $2a(2 - 1) = 6$. Type `6/(2*(2-1))`: it returns 3, so $a = 3$.

3. Then $c = 14 - ab^2$: type `14-3*2`: it returns 8, so $c = 8$.

4. Check: type `y=3*sqrt(2)^x+8` and plot `(2,14)`, `(4,20)` and `(6,32)`; the curve passes through all three.

Answer: 8

20. HARD · DESMOS

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x-4)^2 + 125$. Which of the following statements is true?

(A) Only f attains a maximum, 120

(B) Only g attains a maximum, 125 ✓

(C) Neither attains a maximum

(D) Both f and g attain a maximum

1. Type `y=120-90*0.8^x{x>=0}` and `y=-3(x-4)^2+125{x>=0}`.
2. Tap the top of the parabola: (4, 125), so g attains a maximum of 125.
3. Type `y=120` to draw the line 120: f rises toward it but never touches it (no grey dot appears). For example, type `120-90*0.8^50`: it returns 119.9987, still below 120.
4. So only g attains a maximum, 125.

Answer: (B)

21. HARD · DESMOS

The polynomial $p(x) = x^3 + ax^2 + bx - 6$ is divisible by $x - 2$, and the remainder when $p(x)$ is divided by $x + 1$ is 12. What is the value of $a + b$?

1. $p(2) = 0$ gives $8 + 4a + 2b - 6 = 0$, so $b = -1 - 2a$. $p(-1) = 12$ gives $-1 + a - b - 6 = 12$, so $b = a - 19$. Type `y=-1-2x` and `y=x-19` (letting x stand for a and y for b).
2. Tap the intersection: (6, −13), so $a = 6$ and $b = -13$.
3. Check: type `p(x)=x^3+6x^2-13x-6`, then `p(2)`: it returns 0, and `p(-1)`: it returns 12.
4. Type `6+(-13)`: it returns −7, so $a + b = -7$.

Answer: −7

22. HARD · DESMOS

$$-2(x-5)^2 + k = m(x-1)(x-r)$$

In the equation above, k, m, and r are constants. The equation is true for all real values of x. What is the value of k?

(A) −32

(B) 18

(C) 32 ✓

(D) 50

1. Comparing the x^2 terms gives $m = -2$. At $x = 1$ the right side is 0, so the left side must be 0 there. Type `y=-2(x-5)^2+k` (accept the slider for k) and plot `(1,0)`.
2. Widen the slider range (set its maximum to 40) and slide k until the curve passes through the point: $k = 32$.
3. Check: type `-2*(1-5)^2+32`: it returns 0.
4. Also $r = 9$: type `y=-2(x-1)(x-9)`; it lies exactly on top of `y=-2(x-5)^2+32`.

Answer: (C)

23. HARD · DESMOS

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

(A) 198.4 ✓ (B) 199.2

(C) 187.5 (D) 196.9

1. The doses are Mon 8 a.m., Mon 8 p.m., Tue 8 a.m., Tue 8 p.m., Wed 8 a.m., Wed 8 p.m., Thu 8 a.m.: 7 doses. Each dose is 100 and each earlier dose has been halved once for every dose since.
2. The amount is $100\left(1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{64}\right)=100\cdot\dfrac{1-0.5^7}{1-0.5}$.
3. Type `100*(1-0.5^7)/(1-0.5)` : it returns 198.4375, which is 198.4 to the nearest tenth.

Answer: (A)

24. SAT CHALLENGE · DESMOS

In the xy-plane, the line ℓ touches the parabola $y=x^2-4x+7$ at exactly one point, the point where $x=3$. What is the y-intercept of line ℓ?

1. The line meets the parabola only at $x=3$, so $x^2-4x+7-(mx+b)$ must equal $(x-3)^2=x^2-6x+9$.
2. The x-terms: $-4-m=-6$. Type `-4-(-6)` : it returns 2, so $m=2$.
3. The constants: $7-b=9$. Type `7-9` : it returns -2, so $b=-2$.
4. Check: type `y=x^2-4x+7` and `y=2x-2` ; they touch only at (3, 4).

Answer: -2

25. HARD · DESMOS

The function f is defined by $f(x)=a\cdot b^x$, where a and b are positive constants. For every value of x, $f(x+2)$ is 44% greater than $f(x)$. For every value of x, $f(x+3)$ is p% greater than $f(x)$. What is the value of p?

(A) 132 (B) 60

(C) 66 **(D)** 72.8 ✓

1. $f(x+2)=b^2f(x)$ and it is 44% greater, so $b^2=1.44$. Type `sqrt(1.44)` : it returns 1.2, so $b=1.2$.
2. Now $f(x+3)=b^3f(x)$: type `1.2^3` : it returns 1.728.
3. A factor of 1.728 is an increase of $(1.728-1)\cdot 100$ percent. Type `(1.728-1)*100` : it returns 72.8, so $p=72.8$.

Answer: (D)

26. HARD · DESMOS

The quadratic function f has a zero at $x = 2$. The graph of $y = f(x)$ in the xy-plane passes through the points $(0, -16)$ and $(5, 54)$. What is the minimum value of f?

(A) -9 (B) -1

(C) -18 ✓ (D) -16

1. The points are $(0, -16)$, $(2, 0)$ and $(5, 54)$. Open a table and enter $x_1 = 0, 2, 5$ and $y_1 = -16, 0, 54$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = 2$, $b = 4$, $c = -16$.
2. Type `y=2x^2+4x-16` and tap the vertex (lowest point): $(-1, -18)$.
3. The minimum value of f is -18.

Answer: (C)

27. HARD · DESMOS

The graph of the quadratic function f in the xy-plane passes through the points $(0, 1)$, $(1, -2)$ and $(3, 4)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ 3 units to the right and 5 units down. What is the y-coordinate of the y-intercept of the graph of g?

(A) -1 **(B) 29 ✓**

(C) 34 (D) 39

1. Open a table and enter $x_1 = 0, 1, 3$ and $y_1 = 1, -2, 4$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = 2$, $b = -5$, $c = 1$.
2. Type `f(x)=2x^2-5x+1` and `g(x)=f(x-3)-5`: right 3, down 5.
3. Tap the y-intercept of g: (0, 29). Check: type `g(0)`; it returns 29.

Answer: (B)

28. SAT CHALLENGE · DESMOS

$$\sqrt{x+7} - \sqrt{x-5} = 2$$

What is the solution to the equation above?

1. Type `y=sqrt(x+7)-sqrt(x-5)` and `y=2` (the left side needs $x \geq 5$).
2. Tap the intersection: (9, 2), so $x = 9$.
3. Check: type `sqrt(9+7)-sqrt(9-5)`: it returns 2.

Answer: 9

29. HARD · DESMOS

The function f is linear. The function g is defined by $g(x) = \dfrac{f(x+2) - x}{x - 3}$ for $x \neq 3$. Some values of g are shown in the table below.

x	1	5
$g(x)$	3	-11

What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

(A) -3 (B) -5

(C) -4 **(D) 4 ✓**

1. $g(1) = 3$ gives $\dfrac{f(3) - 1}{1 - 3} = 3$: type `3*(-2)+1`: it returns -5, so $f(3) = -5$. $g(5) = -11$ gives $\dfrac{f(7) - 5}{5 - 3} = -11$: type `-11*2+5`: it returns -17, so $f(7) = -17$.
2. Open a table with $x_1 = 3, 7$ and $y_1 = -5, -17$ and type `y_1~m*x_1+n`: Desmos returns $m = -3$, $n = 4$.
3. The y-intercept is 4. Check with `f(x)=-3x+4`: type `(f(1+2)-1)/(1-3)`; it returns 3, and type `(f(5+2)-5)/(5-3)`; it returns -11.

Answer: (D)

30. HARD · DESMOS

The function f is defined for $x \geq 0$ by

$$f(x) = 48\left(\frac{1}{8}\right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x \geq 0$ and displays the value of $f(0)$ as a constant factor?

(A) $f(x) = 12 \cdot 2^x$ ✓ (B) $f(x) = 48 \cdot 2^x$

(C) $f(x) = 12 \cdot \left(\dfrac{1}{2}\right)^x$ (D) $f(x) = 192 \cdot \left(\dfrac{1}{2}\right)^x$

1. Type `y=48*(1/8)^((2-x)/3)` and tap the y-intercept: (0, 12), so the constant factor must be 12.
2. Type `y=12*2^x`: it lies exactly on top of the first graph.
3. The others fail: `y=192*(1/2)^x` starts at 192, `y=48*2^x` starts at 48, and `y=12*(1/2)^x` starts at 12 but falls instead of rises. Only (A) matches and shows 12.

Answer: (A)

31. HARD · DESMOS

$$\frac{x}{x-2}+\frac{3}{x+2}=\frac{8}{x^2-4}$$

What is the sum of all real solutions of the equation above?

1. Multiply by $(x-2)(x+2)$: $x(x+2)+3(x-2)=8$, so $x^2+5x-14=0$. Type `y=x^2+5x-14` and tap the zeros: $(-7,\,0)$ and $(2,\,0)$.
2. Type `y=x/(x-2)+3/(x+2)-8/(x^2-4)` : its graph has a hole at $x=2$ (the expression is undefined there), and its only x-intercept is $(-7,\,0)$.
3. Check: type `(-7)/(-7-2)+3/(-7+2)-8/((-7)^2-4)` : it returns 0.
4. Only $x=-7$ works, so the sum of all real solutions is -7.

Answer: -7

32. HARD · DESMOS

The function $g(x)=2x^2+bx+c$, where b and c are constants, has zeros $k+1$ and $3k-5$, where k is a constant. If $g(2)=g(10)$, what is the value of c?

(A) 35

(B) 70 ✓

(C) 140

(D) 24

1. $g(2)=g(10)$ puts the axis at $x=6$, so the zeros add to 12: $(k+1)+(3k-5)=12$. Type `y=4x-4` and `y=12` (letting x stand for k).
2. Tap the intersection: $(4,\,12)$, so $k=4$ and the zeros are 5 and 7.
3. Type `y=2(x-5)(x-7)` and tap the y-intercept: $(0,\,70)$, so $c=70$.
4. Check $g(2)=g(10)$: type `2(2-5)(2-7)` and `2(10-5)(10-7)` ; both return 30.

Answer: (B)

33. HARD · DESMOS

Let $g(x)=2^x$ and $h(x)=3x-1$. For which values of x is $g(x)>h(x)$?

(A) $x>3$ only

(B) $x<3$

(C) $x<1$ or $x>3$ ✓

(D) $1<x<3$

1. Type `y=2^x` and `y=3x-1` .
2. Tap the two intersections: $(1,\,2)$ and $(3,\,8)$.
3. The exponential curve is above the line to the left of $x=1$ and to the right of $x=3$, and below it in between. So $g(x)>h(x)$ for $x<1$ or $x>3$.

Answer: (C)

34. HARD · DESMOS

Let $g(x) = 7 + 9\left(\frac{1}{4}\right)^x$ for all real numbers x. Consider the following statements.

I. $g(x) > 7$ for every real number x.
II. The minimum value of g is 7.
III. g is a decreasing function.

Which of the statements must be true?

(A) I and III only ✓

(B) I and II only

(C) II and III only

(D) I, II, and III

1. Type `y=7+9*(1/4)^x` and `y=7`.
2. The curve falls toward the line but never touches it: type `7+9*(1/4)^3`: it returns 7.1406, still above 7. So $g(x) > 7$ for every x (I true), and there is no minimum value (II false).
3. The curve always goes down as x increases, so g is decreasing (III true). I and III only.

Answer: (A)

35. SAT CHALLENGE · DESMOS

The function f is defined by $f(x) = \frac{2x+1}{x-3}$ for all $x \neq 3$, and f^{-1} is the inverse function of f. What is the value of $f^{-1}\left(f^{-1}(5)\right)$?

1. $f^{-1}(5)$ is the input where $f(x) = 5$. Type `y=(2x+1)/(x-3)` and `y=5` and tap the intersection: (5.3333, 5), so $f^{-1}(5) = \frac{16}{3}$.
2. Now find the input where $f(x) = \frac{16}{3}$. Type `y=16/3` and tap the intersection: (5.1, 5.3333).
3. So $f^{-1}(f^{-1}(5)) = 5.1$. Check: type `(2*5.1+1)/(5.1-3)`: it returns 5.3333, which is $\frac{16}{3}$.

Answer: 5.1

36. HARD · DESMOS

The temperature T, in degrees Fahrenheit, of a sample t minutes after it is placed in a chamber is modeled by one of the functions below for $t \geq 0$. For which model is 85 the greatest temperature the sample actually reaches?

(A) $T = 85 - 60(0.8)^t$

(B) $T = 40 + 45(1.05)^t$

(C) $T = 25 + 85(0.8)^t$

(D) $T = 25 + 60(0.8)^t$ ✓

1. Type the four models with x standing for t: `y=25+60*0.8^x`, `y=85-60*0.8^x`, `y=40+45*1.05^x`, `y=25+85*0.8^x`, and the line `y=85`.
2. Tap the y-intercepts: (C) (0, 110), (D) (0, 85), (B) (0, 85), (A) (0, 25).
3. (C) and (D) have factor 0.8, so they fall and their greatest values are the y-intercepts, 110 and 85. (B) has factor 1.05 and rises without limit past 85. (A) rises toward 85 but never reaches it: type `85-60*0.8^30`: it returns 84.9257.
4. Only model (D) has 85 as the greatest temperature actually reached.

Answer: (D)

37. HARD · DESMOS

The quadratic function $h(x) = ax^2 + bx + c$, where a, b and c are constants, satisfies $h(-3) = h(1)$, $h(0) > h(-1)$ and $h(-1) > 0$. Consider the following statements.

I. $b > 0$
II. $c > 0$
III. The equation $h(x) = 0$ has two distinct real solutions.

Which of the statements must be true?

(A) I, II, and III

(B) I only

(C) I and II only ✓

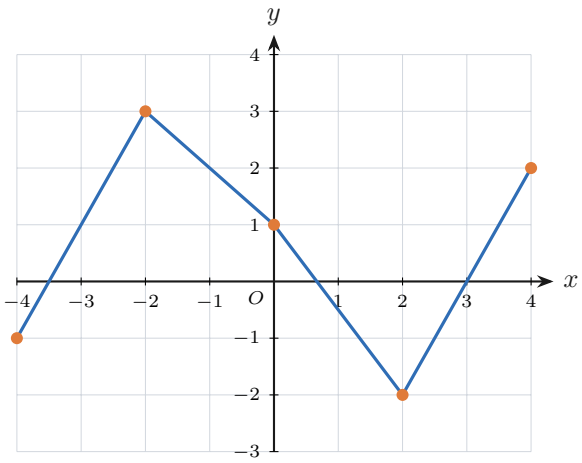
(D) II and III only

1. $h(-3) = h(1)$ puts the axis at $x = -1$. Since $h(0) > h(-1)$, the vertex is the lowest point (opens upward) and $h(-1) > 0$ says it is above the axis. Try `y=(x+1)^2+2`.
2. Tap the vertex: $(-1, 2)$. The curve never meets the x-axis, so III is false. Check $h(-3) = h(1)$: type `(-3+1)^2+2`; it returns 6 and type `(1+1)^2+2`; it returns 6.
3. Type `y=x^2+2x+3`: the same graph, so $b = 2 > 0$ (I) and $c = 3 > 0$ (II). In general $b = 2a > 0$ and $c = h(0) > h(-1) > 0$. I and II only.

Answer: (C)

38. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ is shown for $-4 \le x \le 4$; the function f is not defined for other values of x. The function g is defined by $g(x) = f(x-1) - 1$. For how many values of x is $g(x) = 0$?



(A) 2

(B) 3 ✓

(C) 4

(D) 1

1. $g(x) = 0$ means $f(x-1) = 1$, so find where the graph of f is at height 1.
2. From the graph, $f(u) = 1$ at $u = -3$, $u = 0$ (the point $(0, 1)$) and $u = 3.5$.
3. Since $u = x - 1$, $x = u + 1$: $x = -2$, $x = 1$ and $x = 4.5$. All lie in the domain $-3 \le x \le 5$ of g, so there are 3 values, option (B).

Answer: (B)

39. HARD · DESMOS

The quadratic function f has its vertex at $(3, 4)$. The function g is defined by $g(x) = -2f(x+1) + 7$, and the graph of $y = g(x)$ passes through the point $(4, 2)$. What is the value of $f(9) - f(2)$?

1. $g(4) = -2f(5) + 7 = 2$, so $f(5) = \dfrac{2-7}{-2}$. Type `(2-7)/(-2)` : it returns 2.5.
2. With the vertex $(3, 4)$, $f(x) = a(x-3)^2 + 4$ and $f(5) = 2.5$. Type `(2.5-4)/(5-3)^2` : it returns -0.375, so $a = -0.375$.
3. Type `f(x)=-0.375(x-3)^2+4` , then `f(9)-f(2)` : it returns -13.125, which is $-\dfrac{105}{8}$.
4. Check: type `-2f(5)+7` ; it returns 2.

Answer: $-\dfrac{105}{8}$

40. HARD · DESMOS

A rocket's height, in feet, t seconds after launch is modeled by $h(t) = -16t^2 + bt + c$, where b and c are constants. The rocket is launched from a height of at most 94 feet, and the maximum height it reaches is 130 feet. For how many seconds is the rocket at a height of at least 94 feet?

1. The maximum is 130 at some time v, so $h(t) = -16(t - v)^2 + 130$. Type `y=-16(x-v)^2+130` (accept the slider for v) and `y=94`.
2. Set $v = 3$ and tap the two intersections: (1.5, 94) and (4.5, 94). Change v: the gap between them stays the same.
3. A launch height of at most 94 means $130 - 16v^2 \le 94$, so $v \ge 1.5$ and both crossings come after launch. Type `4.5-1.5`: it returns 3, the seconds spent at or above 94 feet.

Answer: 3

41. HARD · DESMOS

An investment grows exponentially, increasing by 6% every half-year. If V_0 is its initial value and $V(t)$ is its value after t years, after how many years will the value first reach three times the initial value? Round your answer to the nearest hundredth.

1. Every half-year multiplies by 1.06, so after t years the factor is 1.06^{2t}. Type `y=1.06^(2x)` and `y=3` (letting x stand for t).
2. Tap the intersection: (9.4271, 3).
3. Type `ln(3)/(2*ln(1.06))`: it returns 9.4271, which is 9.43 to the nearest hundredth.
4. Check: type `1.06^(2*9.43)`: it returns 3.001, just above 3.

Answer: 9.43

42. HARD · DESMOS

A bakery's daily profit, in dollars, from baking n batches of bread is modeled by $P(n) = -2n^2 + 120n - 500$. The ovens can bake at most 25 batches per day. According to the model, what is the greatest daily profit the bakery can earn?

1. Type `y=-2x^2+120x-500{0<=x<=25}` (letting x stand for n).
2. The vertex of the full parabola is (30, 1300), but $n = 30$ is not allowed. On $0 \le n \le 25$ the curve is still rising at its right end.
3. Type `-2*25^2+120*25-500`: it returns 1250, so the greatest allowed profit is 1250 dollars.

Answer: 1250

43. HARD · DESMOS

The quadratic function $f(x) = 2x^2 + bx + c$ has $(x - 9)$ as a factor. The function g is defined by $g(x) = f(x + 3)$, and the graph of g in the xy-plane has its vertex at $x = 1$. What is the other zero of f?

1. $g(x) = f(x + 3)$ has its vertex at $x = 1$, so the vertex of f is at $x = 4$. The zeros 9 and r are centred on 4: $\dfrac{9 + r}{2} = 4$. Type `y=(9+x)/2` and `y=4` (letting x stand for r).
2. Tap the intersection: $(-1, 4)$, so $r = -1$.
3. Check: type `f(x)=2(x-9)(x+1)` and `g(x)=f(x+3)`, and tap the vertex of g: $(1, -50)$. Its x-value is 1.

Answer: -1

44. HARD · DESMOS

The quadratic function f satisfies $f(-1) = f(7)$, $f(0) = -4$ and $f(4) = 12$. What is the value of $f(10)$?

(A) 0 (B) 98

(C) -98 **(D) -84 ✓**

1. $f(-1) = f(7)$ puts the axis at $x = 3$, so $f(6) = f(0) = -4$. The points are $(0, -4)$, $(6, -4)$, $(4, 12)$.
2. Open a table and enter $x_1 = 0, 6, 4$ and $y_1 = -4, -4, 12$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = -2$, $b = 12$, $c = -4$.
3. Type `-2*10^2+12*10-4`: it returns -84, so $f(10) = -84$.

Answer: (D)

45. SAT CHALLENGE · DESMOS

The function g is defined by $g(x) = |x^2 - 4x|$. Consider the following statements.

I. $g(x) \geq 0$ for every real number x.
II. The graph of $y = g(x)$ has exactly two x-intercepts.
III. g is increasing for every $x > 2$.

Which of the statements must be true?

(A) I and II only ✓ (B) II and III only

(C) I, II, and III (D) I only

1. Type `y=abs(x^2-4x)`.
2. The whole graph is on or above the x-axis (I true), and the x-intercepts are (0, 0) and (4, 0), exactly two (II true).
3. Tap the local maximum (2, 4): after $x = 2$ the graph falls to $(4, 0)$, so g is not increasing for all $x > 2$ (III false). Check: type `abs(3^2-4*3)`: it returns 3 and type `abs(3.5^2-4*3.5)`: it returns 1.75.
4. Only I and II must be true, option (A).

Answer: (A)

46. HARD · DESMOS

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

(A) 785

(B) 839 ✓

(C) 677

(D) 572

1. Open a table and enter $x_1 = 2, 5, 8, 11$ and $y_1 = 44, 95, 200, 359$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = 3$, $b = -4$, $c = 40$.
2. Check: type `y=3x^2-4x+40`; the curve passes through all four table points.
3. Type `3*17^2-4*17+40`: it returns 839, so $g(17) = 839$.

Answer: (B)

47. HARD · DESMOS

The concentration of a medication in the bloodstream, in milligrams per liter, is modeled by $C(t) = 18\left(\frac{1}{3}\right)^{\frac{t-4}{2}}$, where $t \geq 0$ is the number of hours after the first measurement. Which of the following is an equivalent form of $C(t)$ in which the maximum value of C (for $t \geq 0$) appears explicitly as the coefficient of the exponential term?

(A) $486 \cdot 3^{-\frac{t+2}{2}}$

(B) $18 \cdot 3^{\frac{4-t}{2}}$

(C) $162 \cdot 3^{-\frac{t}{2}}$ ✓

(D) $54 \cdot 3^{\frac{2-t}{2}}$

1. Type `y=18*(1/3)^((x-4)/2)` and tap the y-intercept: (0, 162). The curve falls, so its maximum is 162 at $t = 0$.
2. Type each option: `y=486*3^(-(x+2)/2)`, `y=18*3^((4-x)/2)`, `y=162*3^(-x/2)`, `y=54*3^((2-x)/2)`. All four lie exactly on that curve.
3. Only (C) has the maximum value 162 as its coefficient (the others show 486, 18 and 54). Check: type `18*3^2`: it returns 162.

Answer: (C)

48. HARD · DESMOS

A water balloon is launched upward from a rooftop. Its height above the ground, in feet, t seconds after launch is modeled by a quadratic function h. The maximum height of 125 feet is reached at a time between $t = 1$ and $t = 6$, and $h(1) = 80$ and $h(6) = 105$. What is the height, in feet, of the balloon at the moment of launch?

(A) 45 ✓

(B) 75

(C) 37

(D) 80

1. Write $h(t) = 125 - a(t - v)^2$. Then $a(1 - v)^2 = 45$ and $a(6 - v)^2 = 20$. Dividing, $\dfrac{6 - v}{v - 1} = \sqrt{\dfrac{20}{45}}$. Type `sqrt(20/45)` : it returns 0.6667, which is $\dfrac{2}{3}$.

2. Solve $6 - v = \frac{2}{3}(v - 1)$: type `(6+2/3)/(1+2/3)` : it returns 4, so $v = 4$. Then type `45/(1-4)^2` : it returns 5, so $a = 5$.

3. Type `y=125-5(x-4)^2` and plot `(1,80)` and `(6,105)` : it passes through both. Tap the y-intercept: (0, 45).

Answer: (A)

PART B · SELF-ASSESSMENT — 25 PROBLEMS

49. HARD · DESMOS

The function f is defined by $f(x) = x^2 - 4x + 1$. The function g is defined by $g(x) = f(x + 3)$. What are the coordinates of the vertex of the graph of g in the xy-plane?

(A) $(2, 0)$

(B) $(-1, 0)$

(C) $(5, -3)$

(D) $(-1, -3)$ ✓

1. Type `f(x)=x^2-4x+1` and tap the vertex: $(2, -3)$.

2. Type `g(x)=f(x+3)` and tap the vertex: $(-1, -3)$.

3. $f(x + 3)$ moves the graph 3 units left, so the vertex $(2, -3)$ becomes $(-1, -3)$.

Answer: (D)

50. HARD · DESMOS

Selected values of the function g are shown in the table below.

x	1	2	3	4	5
$g(x)$	5	8	13	20	29

Which of the following could define g?

(A) $g(x) = 2^x + 3$

(B) $g(x) = 3x + 2$

(C) $g(x) = x^2 + 4$ ✓

(D) $g(x) = 3 \cdot 2^{x-1} + 2$

1. Open a table and enter $x_1 = 1, 2, 3, 4, 5$ and $y_1 = 5, 8, 13, 20, 29$. Type `y_1~a*x_1^2+b*x_1+c`: Desmos returns $a = 1$, $b = 0$, $c = 4$ with $r^2 = 1$.

2. So $g(x) = x^2 + 4$. Type `y=x^2+4`: the curve passes through all five table points.

3. Check the others: type `3*3+2`: it returns 11 (not 13), and type `2^2+3`: it returns 7 (not 8), and type `3*2^(3-1)+2`: it returns 14 (not 13).

Answer: (C)

51. HARD · DESMOS

The functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2)) - g(f(2))$?

1. Type `f(x)=3x-2` and `g(x)=x^2+1`.

2. Type `f(g(2))`: it returns 13. Type `g(f(2))`: it returns 17.

3. Type `13-17`: it returns -4.

Answer: -4

52. HARD · DESMOS

The value of an investment triples every 5 years. The value grows by the same percent every quarter (every 3 months). Which of the following is closest to that quarterly percent increase?

(A) 10.0%

(B) 5.6% ✓

(C) 6.1%

(D) 24.6%

1. 5 years is 20 quarters, and the value triples over those 20 quarters, so each quarter multiplies the value by $3^{1/20}$.

2. Type `3^(1/20)` : it returns 1.0565.

3. Type `(3^(1/20)-1)*100` : it returns 5.6467, so about 5.6% per quarter.

4. Check: type `1.056^20` : it returns 2.9736, close to 3. (Dividing the yearly rate by 4 would give 6.1%, which compounds to too much.)

Answer: (B)

53. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 6$. Which of the following could be the x-intercepts of the graph of $y = f(2x)$?

(A) $x = 0$ and $x = 8$

(B) $x = -2$ and $x = 6$

(C) $x = -4$ and $x = 12$

(D) $x = -1$ and $x = 3$ ✓

1. The graph of $y = f(2x)$ has an x-intercept wherever $2x$ is an x-intercept of f, so $2x = -2$ or $2x = 6$.

2. Solve each: $x = \dfrac{-2}{2} = -1$ and $x = \dfrac{6}{2} = 3$.

3. Replacing x by $2x$ squeezes the graph horizontally by a factor of 2, so each x-intercept is halved: $x = -1$ and $x = 3$.

Answer: (D)

54. HARD · DESMOS

A culture starts with 200 cells. The number of cells triples every 40 minutes. After how many hours will the culture contain exactly 16,200 cells? Enter your answer as a fraction.

1. Let x be the number of tripling periods: type `y=200*3^x` and `y=16200` .

2. Tap the intersection: (4, 16200), so there are 4 periods.

3. Type `4*40` : it returns 160 minutes, and type `160/60` : it returns 2.6667, which is $\dfrac{8}{3}$ hours.

Answer: $\dfrac{8}{3}$

55. HARD · NO CALCULATOR NEEDED

The quadratic function f has a minimum value of -5 at $x = 4$. The function g is defined by $g(x) = -2f(x-3)+1$. What is the maximum value of g?

1. Shifting the graph 3 units right does not change its lowest value, so $f(x-3)$ still has a minimum of -5.

2. Multiplying by -2 flips the graph over, so the minimum becomes a maximum: $-2 \cdot (-5) = 10$.

3. Adding 1 raises the graph by 1: $10 + 1 = 11$, the maximum value of g.

Answer: 11

56. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 2|x-2| - 3$. What is the sum of the x-intercepts of the graph of $y = f(x)$ in the xy-plane?

(A) 4 ✓ (B) $\frac{7}{2}$

(C) 5 (D) 3

1. Set $2|x-2| - 3 = 0$, so $|x-2| = \frac{3}{2}$.

2. Then $x - 2 = \frac{3}{2}$ or $x - 2 = -\frac{3}{2}$, so $x = \frac{7}{2}$ or $x = \frac{1}{2}$.

3. The sum is $\frac{7}{2} + \frac{1}{2} = 4$, which is twice the vertex x-coordinate 2. That is option (A).

Answer: (A)

57. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = |x-3|$ and the function g is defined by $g(x) = f(x+1) - 2$. What are the coordinates of the vertex of the graph of g?

(A) $(4, -2)$ **(B)** $(2, -2)$ ✓

(C) $(2, 2)$ (D) $(4, 2)$

1. The vertex of $f(x) = |x-3|$ is $(3, 0)$.

2. $f(x+1)$ moves the graph 1 unit left, to $(2, 0)$, and the -2 moves it 2 units down, to $(2, -2)$.

3. Check by rewriting: $g(x) = |(x+1) - 3| - 2 = |x-2| - 2$, whose vertex is $(2, -2)$.

Answer: (B)

58. HARD · NO CALCULATOR NEEDED

$$\frac{6}{x-3}+2=\frac{x+3}{x-3}$$

Which of the following describes the real solutions of the equation above?

(A) There are exactly two real solutions

(B) There are infinitely many real solutions

(C) There are no real solutions ✓

(D) There is exactly one real solution

1. The expressions are undefined at $x=3$. Multiply both sides by $x-3$: $6+2(x-3)=x+3$.
2. Simplify: $2x=x+3$, so $x=3$.
3. But $x=3$ makes the denominators zero, so it is extraneous. The equation has no real solutions, option (C).
Answer: (C)

59. HARD · DESMOS

The function f is defined by $f(x)=2x+1$. The function g is defined by $g(x)=f(f(x))-f(x)$. What is the value of $g(3)$?

1. Type `f(x)=2x+1` and `g(x)=f(f(x))-f(x)`.
2. Type `g(3)`: it returns 8.
3. Check by hand: $f(3)=7$ and $f(f(3))=f(7)=15$, and $15-7=8$.
Answer: 8

60. HARD · DESMOS

The function f is defined by $f(x)=(x-1)^2+4$. The function g is defined by $g(x)=f(x-k)+m$, where k and m are constants. The vertex of the graph of g is $(6,-3)$. What is the value of $k+m$?

1. Type `f(x)=(x-1)^2+4` and tap its vertex: (1, 4).
2. The vertex moves from $(1,4)$ to $(6,-3)$. Horizontally: type `6-1`: it returns 5, so $k=5$. Vertically: type `-3-4`: it returns -7, so $m=-7$.
3. Type `g(x)=f(x-5)-7` and tap the vertex: (6, −3).
4. Type `5+(-7)`: it returns -2, so $k+m=-2$.
Answer: -2

61. HARD · DESMOS

The function f is defined by $f(x) = x^2 - 2x$. For how many real values of x is $f(f(x)) = 0$?

(A) 4 ✓ (B) 1

(C) 2 (D) 3

1. Type `f(x)=x^2-2x` and `y=f(f(x))`.

2. Tap the x-intercepts: $(-0.7321, 0)$, $(0, 0)$, $(2, 0)$ and $(2.7321, 0)$: four of them.

3. Check by hand: $f(x) = 0$ gives $x = 0, 2$; $f(x) = 2$ gives $x = 1 \pm \sqrt{3}$. Type `1-sqrt(3)`: it returns -0.7321 and type `1+sqrt(3)`: it returns 2.7321.

4. So there are 4 values, option (A).

Answer: (A)

62. HARD · DESMOS

The functions f and g are defined by $f(x) = x + 3$ and $g(x) = 2x - 1$. For which values of x is $f(g(x)) = g(f(x))$?

(A) Only $x = 0$ (B) Only $x = 3$

(C) All real numbers **(D) No real numbers ✓**

1. Type `f(x)=x+3` and `g(x)=2x-1`.

2. Type `y=f(g(x))` and `y=g(f(x))`: two lines, $y = 2x + 2$ and $y = 2x + 5$.

3. They are parallel, so there is no intersection and no x with $f(g(x)) = g(f(x))$.

Answer: (D)

63. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane passes through the point $(6, -2)$. The function g is defined by $g(x) = 2f(3 - x) + 1$. Which of the following points must lie on the graph of $y = g(x)$?

(A) $(3, -3)$ (B) $(-3, -5)$

(C) $(-3, -3)$ ✓ (D) $(9, -3)$

1. $g(x) = 2f(3 - x) + 1$ uses f at the input $3 - x$. The known point has $f(6) = -2$, so we need $3 - x = 6$.

2. That gives $x = -3$. Then $g(-3) = 2f(6) + 1 = 2(-2) + 1 = -3$.

3. So $(-3, -3)$ must lie on the graph of g, option (C). The x-value is not $6 + 3$ or $6 - 3$ because x is subtracted inside.

Answer: (C)

64. HARD · NO CALCULATOR NEEDED

The function f satisfies $f(1) = 3$ and $f(3) = 9$. The functions g and h are defined by $g(x) = f(x) + 4$ and $h(x) = f(x - 2)$. What is the value of $g(3) + h(3)$?

1. $g(3) = f(3) + 4 = 9 + 4 = 13$.

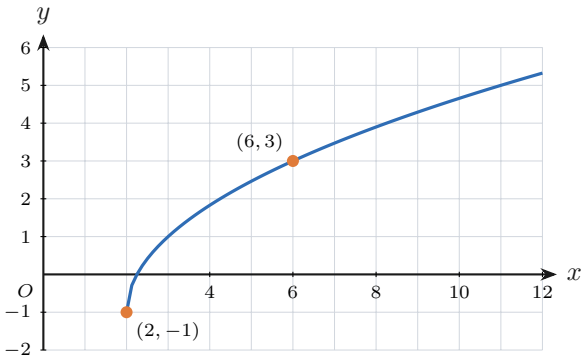
2. $h(3) = f(3 - 2) = f(1) = 3$.

3. $g(3) + h(3) = 13 + 3 = 16$.

Answer: 16

65. HARD · DESMOS

The graph of the function $f(x) = a\sqrt{x - h} + k$ is shown, where a, h and k are constants. The graph starts at the point $(2, -1)$ and passes through the point $(6, 3)$. What is the value of $f(11)$?



1. The graph starts at $(h, k) = (2, -1)$. Type `f(x)=a*sqrt(x-2)-1` (accept the slider for a) and plot `(6,3)`.

2. Slide a until the curve passes through the point: $a = 2$.

3. Type `f(11)` : it returns 5.

Answer: 5

66. HARD · NO CALCULATOR NEEDED

The graph of $y = f(x)$ in the xy-plane has its vertex at $(-2, 5)$. What are the coordinates of the vertex of the graph of $y = -f(x + 1) + 3$?

(A) $(-1, 8)$ **(B) $(-3, -2)$ ✓**

(C) $(-1, -2)$ (D) $(-3, 8)$

1. $f(x + 1)$ moves the vertex 1 unit left: $(-2, 5) \to (-3, 5)$.
2. The minus sign reflects the graph over the x-axis, so $(-3, 5) \to (-3, -5)$.
3. The $+3$ moves it up 3: $(-3, -5) \to (-3, -2)$.

Answer: (B)

67. HARD · DESMOS

A store sells $300 - 4p$ units of a product per week when the price is p dollars. The store only sets prices that are multiples of 5 dollars. What is the greatest weekly revenue, in dollars, the store can earn?

(A) 5,500 (B) 5,400

(C) 5,625 **(D) 5,600 ✓**

1. Type `y=x(300-4x)` (letting x stand for p). The vertex is (37.5, 5625), but 37.5 is not a multiple of 5.
2. Try the nearest multiples of 5: type `35*(300-4*35)` : it returns 5600 and type `40*(300-4*40)` : it returns 5600.
3. Farther prices give less: type `30*(300-4*30)` : it returns 5400, and type `45*(300-4*45)` : it returns 5400. The greatest allowed revenue is 5,600 dollars, option (D).

Answer: (D)

68. HARD · DESMOS

The graph of $g(x) = (x - h)^2 + k$ in the xy-plane has a minimum value of 2 and passes through the points $(1, 6)$ and $(5, 6)$. What is the value of $h + k$?

1. The minimum value is 2, so $k = 2$. The points $(1, 6)$ and $(5, 6)$ are at the same height, so the axis is halfway: type `(1+5)/2` : it returns 3, so $h = 3$.
2. Type `y=(x-3)^2+2` and plot `(1,6)` and `(5,6)` : the curve passes through both. Tap the vertex: (3, 2).
3. Type `3+2` : it returns 5, so $h + k = 5$.

Answer: 5

69. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = x^4 - 13x^2 + 36$. Which of the following is an equivalent form of $f(x)$ in which the zeros of f appear as constants?

(A) $f(x) = (x-2)(x+2)(x-3)(x+3)$ ✓

(B) $f(x) = (x^2-4)(x^2-9)$

(C) $f(x) = (x-1)(x+1)(x-6)(x+6)$

(D) $f(x) = (x-2)^2(x+3)^2$

1. Treat $x^4 - 13x^2 + 36$ as a quadratic in x^2: $(x^2-4)(x^2-9)$.
2. Factor each difference of squares: $(x-2)(x+2)(x-3)(x+3)$.
3. The zeros -2, 2, -3, 3 now appear as constants, option (A). The form $(x^2-4)(x^2-9)$ is equivalent but still has x^2 in each factor.

Answer: (A)

70. HARD · DESMOS

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c, and $x+3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c?

1. $p(-3) = 0$ and the constant term is 12, so $p(x) = (x+3)(x^2+ux+4)$ with u an integer. For one real zero the quadratic factor must have none. Type `y=x^2+a*x+4` (accept the slider for a, in the place of u).
2. Slide a: the parabola stays above the x-axis for $a = -3, \ldots, 3$ and touches it at $a = \pm 4$ (a new zero). So the largest allowed a is 3.
3. Expanding, $c = 3a + 4$: type `3*3+4` : it returns 13.
4. Check: type `y=x^3+6x^2+13x+12` ; it crosses the x-axis only at $(-3, 0)$.

Answer: 13

71. HARD · NO CALCULATOR NEEDED

For all real numbers x, $4 \cdot 8^{(2x-1)/3} = a \cdot b^x$, where a and b are positive constants. What is the value of $a + b$?

1. Write $8 = 2^3$: $8^{\frac{2x-1}{3}} = 2^{2x-1}$.
2. Then $4 \cdot 2^{2x-1} = 2^2 \cdot 2^{-1} \cdot 2^{2x} = 2 \cdot 4^x$.
3. So $a = 2$ and $b = 4$, and $a + b = 6$.

Answer: 6

72. Medium · Desmos

$$57x^2 + (57b + a)x + ab = 0$$

In the equation above, a and b are positive constants. The product of the solutions is kab, where k is a constant. What is the value of k? Enter your answer as a fraction.

1. Try $a = 3$ and $b = 2$: type `y=57x^2+(57*2+3)x+3*2`.

2. Tap the zeros: $(-2, 0)$ and $(-0.0526, 0)$.

3. Their product over ab: type `(-2)*(-3/57)/6`: it returns 0.0175.

4. Type `1/57`: it returns 0.0175, so $k = \frac{1}{57}$.

Answer: $\frac{1}{57}$

73. Hard · Desmos

$$(ax + 3)(5x^2 - bx + 4) = 20x^3 - 9x^2 - 2x + 12$$

In the equation above, a and b are constants. What is the value of ab?

(A) 20 **(B) 24 ✓**

(C) 40 (D) 18

1. The x^3 terms: $5a = 20$. Type `20/5`: it returns 4, so $a = 4$.

2. The x^2 terms: $-ab + 15 = -9$. Type `15-(-9)`: it returns 24, so $ab = 24$.

3. Check with $b = 6$: type `24/4`: it returns 6. Type `y=(4x+3)(5x^2-6x+4)` and `y=20x^3-9x^2-2x+12`; the two curves coincide.

Answer: (B)